



Eksamen 2020, spørgsmål

Introduction to future energy (Danmarks Tekniske Universitet)

▼ Lecture 1 (Introduction)

restart : with(LinearAlgebra) :

Q 1-1) What (roughly) is the percentage of global energy supplied by oil?

☐ A) 18%

☐ B) 23%

☐ C) 28%

☒ D) 33%

☐ E) 40%

Q 1-2) How much power (roughly) does the world economy consume?

☐ A) 1.700 GW

☒ B) 17.000 GW

☐ C) 27.000 GW

☐ D) 43.000 GW

☐ E) 100 TW

Q 1-3) Why do metals produced as bi-products typically have lower availability for large-scale usage compared those produced as primary products?

☐ A) Because they have lower crustal abundance

☒ B) Because they are inelastic to price changes

☐ C) Because they are elastic to price changes

☐ D) Because they are more expensive

☐ E) Because they are poisonous to the environment

Q 1-4) What is the main observation called "the demographic transition"?

- ☐ A) That poor countries grow populations very fast
- ☐ B) That rich countries help poor countries to obtain a good standard of living
- ☐ C) That the rich get richer at the expense of the poor
- ☐ D) That the number of childbirths/women increases with improved health care and education
- ☒ E) That the number of childbirths/woman decreases with improved health care and education

Question 1

Q 1-1) What happens to a country in the middle stage of the demographic transition?

- ☒ A) Population grows because the birth rate is high and the death rate is low
- ☐ B) Population grows fast because the country becomes more attractive to live in and people immigrate in
- ☐ C) Population decreases because death rates rise steeply
- ☐ D) Population decreases because people immigrate out of the country because it becomes less attractive to live in
- ☐ E) Population is stable because birth rates and death rates are equal

Question 2

Q 1-2) At roughly what annual rate has global energy consumption been growing in the past decade?

- ☐ A) -0.2%
- ☐ B) 0.3%
- ☐ C) 0.9%
- ☒ D) 1.8%
- ☐ E) 3.1%

Question 3

Q 1-3) How many square kilometres would a photovoltaic park giving 1 TW of annual average power be (roughly)?

- ☐ A) 3,000 km² (the size of the Danish island of Fyn)
- ☒ B) 50,000 km² (the size of the of Costa Rica)
- ☐ C) 140,000 km² (the size of the island of Java)
- ☐ D) 310,000 km² (the size of Poland)
- ☐ E) 590,000 km² (the size of Madagascar)

Question 4

Q 1-4) Roughly, what proportion of world energy is supplied by natural gas?

- ☐ A) 5%
- ☐ B) 16%
- ☒ C) 23%
- ☐ D) 33%
- ☐ E) 45%

Spørgsmål 1

Q 1-1) Denmark consumes around 4 GW of electricity on average. Assume that we build a photovoltaic plant in Denmark to provide 4 GW of average power (annual average). How big (roughly) would it be?

- ☐ A) About 1/2 km² (the size of DTU Lyngby campus)
- ☐ B) About 5 km² (the size of Strynø island)
- ☐ C) About 17 km² (the size of Saltholm island)
- ☐ D) About 95 km² (the size of Amager island)
- ☒ E) About 350 km² (the size of Mors island)

Spørgsmål 2

Q 1-2) China is the World's most energy consuming country - what percentage of the global energy use happens in China?

- ☐ A) About 12%
- ☒ B) About 22%
- ☐ C) About 32%
- ☐ D) About 42%
- ☐ E) About 52%

Spørgsmål 3

Q 1-3) What does it mean that metal A has an inelastic supply compared to metal B.

- ☐ A) That metal A is more expensive (more \$/kg) than metal B
- ☐ B) That metal A is only produced in a few places in the World, whereas metal B is produced more places
- ☐ C) That metal A has a higher elastic modulus (Young's modulus) than metal B
- ☒ D) That the production of metal A responds less strongly than that of metal B to a change in the market price of the metals
- ☐ E) That metal B is a bi-product of metal A

▼ Lecture 2 (Wind)

restart : with(LinearAlgebra) :

Q 6-1) What is the main degree of freedom of the wind turbine to measure the blade position during the operation.

- ☐ A) Yaw
- ☐ B) Pitch
- ☒ C) Azimuth
- ☐ D) Tilt
- ☐ E) Cone

Q 6-2) There are list of wind turbines. Please select the rotating drag-based machine

- ☐ A) Three-bladed horizontal axis wind turbine
- ☐ B) Two-bladed horizontal axis wind turbine
- ☐ C) Darrieus type wind turbine
- ☐ D) Three-bladed horizontal axis turbine with diffuser
- ☒ E) Savonius type wind turbine

There is an artificial 5MW wind turbine. Its characteristics are shown below. Rotor diameter: 126m; Hub height: 90m; Wind class: IA, $V_{ref} = 50\text{m/s}$, $I_{ref} = 0.16$; Thrust is provided below.

Wind speed (m/s)	Thrust (kN)
4	84
5	131
6	189
7	257
8	336
9	425
10	525
11	411
12	353
13	315
14	286

Q 6-3) Please calculate the minimum, mean and maximum blade root flapwise bending moments at 12m/s

- ☐ A) -4869kNm, 4942kNm, 14753kNm (minimum, mean, maximum)
- ☐ B) -3553kNm, 4942kNm, 15061kNm (minimum, mean, maximum)
- ☐ C) -5905kNm, 4942kNm, 14725kNm (minimum, mean, maximum)
- ☐ D) -1453kNm, 4942kNm, 16153kNm (minimum, mean, maximum)
- ☐ E) -1810kNm, 1834kNm, 5478kNm (minimum, mean, maximum)

Q 6-4) Please calculate the minimum, mean and maximum tower bottom bending moments at 14m/s

- ☐ A) -22842kNm, 36990kNm, 96822kNm (minimum, mean, maximum)
- ☐ B) -15102kNm, 25740kNm, 91602kNm (minimum, mean, maximum)
- ☐ C) -37692kNm, 25740kNm, 94662kNm (minimum, mean, maximum)
- ☐ D) -43812kNm, 25740kNm, 95292kNm (minimum, mean, maximum)
- ☐ E) -31302kNm, 25740kNm, 94842kNm (minimum, mean, maximum)

Question 17

Q 5-1) What is the instant wind power density per flow area (W/m^2) if the wind speed is $v = 10 \text{ m/s}$ at 100 m height and the air density is $\rho = 1.225 \text{ kg/m}^3$?

- ☐ A) 300-400 W/m^2
- ☐ B) 400-500 W/m^2
- ☐ C) 500-600 W/m^2
- ☐ D) 600-700 W/m^2
- ☐ E) 700-800 W/m^2

Question 18

Q 5-2) If the wind speed is measured over a year at a site and found to be described by the Weibull distribution with a shape parameter $k = 2$ and an average wind speed of $v_{ave} = 8.5 \text{ m/s}$. Which wind class should then be used for choosing a turbine for installation at that site?

- ☐ A) Class 0
- ☐ B) Class I
- ☐ C) Class II
- ☐ D) Class III
- ☐ E) Class IV

Question 19

Q 5-3) How many rotational axis, that can be rotated, does a modern horizontal axis individual blade pitch regulated wind turbine have?

- ☐ A) 2
- ☐ B) 3
- ☐ C) 4
- ☐ D) 5
- ☐ E) 6

Question 20

Q 5-4) If a 10 MW wind turbine is placed offshore and have a Capacity Factor of $c = 0.50$. What is the Annual Energy Production (AEP) in $[\text{GWh/year}]$?

- ☐ A) 30-40 GWh/year
- ☐ B) 40-50 GWh/year
- ☐ C) 50-60 GWh/year
- ☐ D) 60-70 GWh/year
- ☐ E) 70-80 GWh/year

Spørgsmål 36

Q 10-1) When is it necessary to change the yaw angle of a wind turbine?

- ☐ A) When the wind speed is changing
- ☐ B) When the air temperature is changing
- ☐ C) When the turbulence is changing
- ☐ D) When the wind direction is changing
- ☐ E) When the grid voltage is changing

Spørgsmål 37

Q 10-2) The world's largest wind turbine has a rotor diameter of $D = 220$ m. What is the maximum available power of the wind if the turbine is placed in a wind speed of $v = 10$ m/s at 150 m height and the air density is $\rho = 1.225$ kg/m³?

- ☐ A) 1-5 MW
- ☐ B) 5-10 MW
- ☐ C) 10-20 MW
- ☐ D) 20-30 MW
- ☐ E) 30-40 MW

Spørgsmål 38

Q 10-3) If the wind speed is measured over many years at a site and found to be described by the Weibull distribution with a shape parameter $k = 2$ and an average wind speed of $v_{ave} = 10.1$ m/s. Which wind class should then be used for choosing a turbine for installation at that site?

- ☐ A) IEC Class 0
- ☐ B) IEC Class 1
- ☐ C) IEC Class 2
- ☐ D) IEC Class 3
- ☐ E) IEC Class 4

Spørgsmål 39

Q 10-4) The worlds largest wind turbine with a power rating of 12 MW has been claimed to produce an Annual Energy Production (AEP) of 67 GWh at an IEC wind class I. What is the corresponding Capacity Factor of the turbine ?

- ☐ A) 20 - 30 %
- ☐ B) 30 - 40 %
- ☐ C) 40 - 50 %
- ☐ D) 50 - 60 %
- ☐ E) 60 - 70 %

Lecture 3 (Solar)

restart : with(LinearAlgebra) :

Q 7-1) What is the main advantage of CSP over PV?

- ☐ A) That it is cheaper (lower \$/W)
- ☐ B) That it can run in more geographical locations
- ☐ C) That it is faster to build
- ☐ D) That it is cheaper to maintain
- ☐ E) That it allows energy storage

Q 7-2) The optimal band-gap of a solar cell is a compromise between

- ☐ A) The cell current and the cell cost
- ☐ B) The cell current and the cell voltage
- ☐ C) The cell current and the cell fill-factor (FF)
- ☐ D) The cell voltage and ambient temperature
- ☐ E) The cell voltage and the fill-factor (FF)

Q 7-3) If you wish to run an electrochemical process and you need a voltage of more than 1 volt (but less than 2 volts) a tandem cell is a much better choice than a single-bandgap cell. Why?

- ☐ A) Because the single photon cell cannot provide sufficient voltage
- ☐ B) Because the single photon cell has a bad fill factor
- ☐ C) Because the tandem cell provides more current at the desired voltage
- ☐ D) Because the tandem cell is a simpler design
- ☐ E) Because the tandem cell is less affected by changes in incident spectrum

Q 7-4) If you build identical photovoltaic plants near Copenhagen and near Rome, you get xx % more annual production in Rome.

- ☐ A) xx = ~0 %
- ☐ B) xx = ~15 %
- ☐ C) xx = ~50 %
- ☐ D) xx = ~80 %
- ☐ E) xx = ~110 %

Question 9

Q 3-1) Our civilization takes around 18 TW of power to run. The earth is hit by much more solar power, but how much more?

- ☐ A) 20 times more
- ☐ B) 500 times more
- ☐ C) 6,500 times more
- ☐ D) 30,000 times more
- ☐ E) 120,000 times more

Question 10

Q 3-2) What has been the approximate annual growth rate of global photovoltaic capacity (until 2017)

- ☐ A) 15%
- ☐ B) 35%
- ☐ C) 65%
- ☐ D) 90%
- ☐ E) 105%

Question 11

Q 3-3) The bandgap is a central concept of a photovoltaic cell. What is it?

- ☐ A) It is the minimum energy which a photon must have to enter the material (else, it is reflected)
- ☐ B) It is the minimum energy which a photon must have to be absorbed (to generate electron-hole pairs)
- ☐ C) It is the maximum energy which a photon can have in order not to generate "hot electrons" in the material
- ☐ D) It is a physical gap (void) in the cell structure between the n-type layer and the p-type layer
- ☐ E) It is a band of vertically oriented gaps in the cell surface (anti-refelction layer)

Question 12

Q 3-4) If you install a solar collector on a well-oriented roof in Lyngby with a 1 kW peak power (i.e.e 1000 W nameplate power). What is roughly, the annual total electricity production you would expect this to generate?

- ☐ A) ca 24 kWh
- ☐ B) ca 50 kWh
- ☐ C) ca 88 kWh
- ☐ D) ca 350 kWh
- ☐ E) ca 950 kWh

Spørgsmål 8

Q 3-1) If you make a photovoltaic installation of a given size how much larger would its annual total production (i.e. # of kWh) be if you build near Rome instead of near Copenhagen?

- ☐ A) About the same
- ☐ B) About 5% larger output
- ☐ C) About 50% larger output
- ☐ D) About 120% larger output
- ☐ E) About 270% larger output

Spørgsmål 9

Q 3-2) Why can tandem photoabsorbers make much more efficient solar cells than single absorber designs?

- ☐ A) Because they are simpler and fewer losses
- ☐ B) Because they get more than twice the voltage, but half the current
- ☐ C) Because their sensitivity to clouds and other non-ideal illumination is less pronounced
- ☐ D) Because they get more than twice the current, but half the voltage
- ☐ E) Because they get more than twice the current, and about the same voltage

Spørgsmål 10

Q 3-3) If you are looking for a good top-cell absorber material to use together with a silicon-based bottom cell in a photovoltaic tandem, the band-gap energy of the top-cell absorber should be in the range:

- ☐ A) 2.5 eV to 3.0 eV
- ☐ B) 1.7 eV to 1.9 eV
- ☐ C) 1.1 eV to 1.3 eV
- ☐ D) 0.3 eV to 0.5 eV
- ☐ E) -1.1 eV to 0 eV

Spørgsmål 11

Q 3-4) What is the principal advantage which Concentrated Solar Power can have over photovoltaics in generating electricity?

- ☐ A) Optional storage for when the sun doesn't shine
- ☐ B) Better scalability of the plant size
- ☐ C) Much higher solar-to-electricity efficiency
- ☐ D) Lower cost (less \$/Watt)
- ☐ E) Better ability to handle non-ideal weather

▼ Lecture 4 (Biomass)

restart : with(LinearAlgebra) :

Q 4-1) What type of renewable energy resource has the largest energy input to the Danish energy system?

- ☐ A) Wind
- ☐ B) Solar
- ☐ C) Hydro
- ☐ D) Geothermal
- ☐ E) Biomass

Q 4-2) What characteristic is not important for biomass gasification for fuel synthesis?

- ☐ A) A high district heating production
- ☐ B) A high output of CO and H₂
- ☐ C) A low content of tars in the output gas
- ☐ D) A high energy efficiency
- ☐ E) Biomass fuel flexibility

Q 4-3) What is the energy efficiency (Lower Heating Value, LHV) of the chemical reaction converting CO₂ and H₂ to CH₃OH (methanol) by $\text{CO}_2 + 3\text{H}_2 \rightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$?

- ☐ A) 80%
- ☐ B) 83%
- ☐ C) 85%
- ☐ D) 88%
- ☐ E) 90%

Q 4-4) By integrating hydrogen from water electrolysis, the biofuel production will be limited by the carbon content of the biomass. Calculate the potential output of DME per kg of biomass input. The biomass carbon content can be assumed to be 45 wt%.

- ☐ A) 0.63 kg
- ☐ B) 0.75 kg
- ☐ C) 0.86 kg
- ☐ D) 1 kg
- ☐ E) 1.26 kg

Question 5

Q 2-1) Concerning thermal gasification. What is not true?

- ☐ A) It can convert a solid fuel to syngas
- ☐ B) It can convert lignin to gas
- ☐ C) It produces a biogas containing mainly methane
- ☐ D) It has high energy efficiency (75-93%)
- ☐ E) Heat is required for the chemical reaction producing gas from char and steam

Question 6

Q 2-2) What is the synergy between thermal gasification and electrolysis?

- ☐ A) Gasification provides heat for electrolysis
- ☐ B) Electrolysis provides heat for gasification
- ☐ C) The gasifier produces gas for electrolysis
- ☐ D) Electrolysis provides oxygen for gasification and hydrogen for syngas production
- ☐ E) Gasification provides oxygen for electrolysis

Question 7

Q 2-3) Calculate the conversion energy efficiency of the chemical reaction $4\text{H}_2 + \text{CO}_2 \rightarrow \text{CH}_4 + 2\text{H}_2\text{O}$ given the following heating values: $\text{LHV}_{\text{H}_2} = 241.8 \text{ MJ/kmol}$, $\text{LHV}_{\text{CH}_4} = 802.3 \text{ MJ/kmol}$

- ☐ A) 85.1%
- ☐ B) 80.0%
- ☐ C) 83.0%
- ☐ D) 78.2%
- ☐ E) 100%

Question 8

Q 2-4) If the production of biofuel is only limited by the carbon content, what is the potential output of ethanol per kg of biomass input. The carbon content of biomass can be assumed to be 47 wt%.

- ☐ A) 0.90 kg
- ☐ B) 1.00 kg
- ☐ C) 1.11 kg
- ☐ D) 0.85 kg
- ☐ E) 0.63 kg

Spørgsmål 4

Q 2-1) What is the approx. use of biomass in the Danish energy system per year?

☐ A) 100 PJ

☐ B) 60 TJ

☐ C) 140 TJ

☐ D) 140 PJ

☐ E) 60 PJ

Spørgsmål 5

Q 2-2) How much hydrogen from electrolysis can approx. be incorporated in biofuel production based on gasification when assuming a consumption of biomass of 10 GJ?

☐ A) 8 GJ

☐ B) 11 GJ

☐ C) 5 GJ

☐ D) 2 GJ

☐ E) 19 GJ

Spørgsmål 6

Q 2-3) Calculate the energy efficiency of gasification if the input is 100 MW of biomass and the output is 0.4 kmol/s of syngas containing 40 mol% H₂ and 40 mol% of CO. LHV_H₂ = 241.8 MJ/kmol, LHV_CO = 283 MJ/kmol

☐ A) 84%

☐ B) 92%

☐ C) 100%

☐ D) 55%

☐ E) 77%

Spørgsmål 7

Q 2-4) How much heat is released in the water gas shift reaction ($\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$) if 0.1 kmol/s of CO is converted? $\text{LHV}_{\text{H}_2} = 241.8 \text{ MJ/kmol}$, $\text{LHV}_{\text{CO}} = 283 \text{ MJ/kmol}$.

- ☐ A) 4 MW
- ☐ B) 4 kW
- ☐ C) 7 MW
- ☐ D) 7 kW
- ☐ E) 24 MW

Lecture 5 (Nuclear)

restart : with(LinearAlgebra) :

Q 8-1) What is the nuclear fission-share of the European (EU) electricity generation?

- ☐ A) ~25%
- ☐ B) ~35%
- ☐ C) ~45%
- ☐ D) ~55%
- ☐ E) ~65%

Q 8-2) What is the approximate energy utilization of natural uranium in a LWR once-through fuel cycle?

- ☐ A) 1%
- ☐ B) 5%
- ☐ C) 15%
- ☐ D) 25%
- ☐ E) 35%

Q 8-3) These nuclides are fissile

- ☐ A) ^{233}U , ^{235}U , ^{238}U
- ☐ B) ^{233}U , ^{235}U , ^{239}Pu
- ☐ C) ^{238}U , ^{239}Pu , ^{232}Th
- ☐ D) ^{233}U , ^{238}U , ^{232}Th
- ☐ E) ^{235}U , ^{238}U , ^{239}Pu

Q 8-4) Nuclear fusion is based on the interactions of

- ☐ A) D+T
- ☐ B) D+U
- ☐ C) T+U
- ☐ D) U+n
- ☐ E) Th+n

Question 13

Q 4-1) What is the nuclear fission share of the global electricity generation?

- ☐ A) 5%
- ☐ B) 10%
- ☐ C) 15%
- ☐ D) 20%
- ☐ E) 25%

Question 14

Q 4-2) What is the approximate energy gain using breeder reactors and a closed fuel cycle, compared to a LWR once-through fuel cycle?

- ☐ A) 20%
- ☐ B) 100%
- ☐ C) 500%
- ☐ D) 2000%
- ☐ E) 10000%

Question 15

Q 4-3) In a fast reactor, the term "fast" refers to

- ☐ A) Construction time
- ☐ B) Neutron spectrum
- ☐ C) Fission fragment energy
- ☐ D) Load-following capacity
- ☐ E) Ability to use thorium fuel

Question 16

Q 4-4) In a reactor, thorium fuel is transformed to ^{233}U by

- ☐ A) Neutron capture
- ☐ B) Double beta-decay
- ☐ C) Neutron capture and double beta-decay
- ☐ D) Fission
- ☐ E) Neutron capture and fission

Spørgsmål 32

Q 9-1) What is the nuclear fission share of the electricity generation in France?

- ☐ 5%
- ☐ 15%
- ☐ 35%
- ☐ 50%
- ☐ 75%

Spørgsmål 33

Q 9-2) A country **without** nuclear power plants

- ☐ Finland
- ☐ Sweden
- ☐ Poland
- ☐ Ukraine
- ☐ Hungary

Spørgsmål 34

Q 9-3) In fission reactors, plutonium is the result of

- ☐ By-product from uranium mining
- ☐ Fission of natural uranium
- ☐ Neutron capture in natural uranium
- ☐ Radioactive decay of natural uranium
- ☐ Radioactive decay of enriched uranium

Spørgsmål 35

Q 9-4) In a "thermal" fission reactor, the term "thermal" refers to

- ☐ Reactor temperature
- ☐ Water coolant temperature
- ☐ Any reactor using liquid coolant
- ☐ The neutron spectrum
- ☐ The fission fragment energy

Lecture 6 (Water)

restart : with(LinearAlgebra) :

Q 5-1) The definition of "celerity" of a linear deep-water wave is:

- ☐ A) The speed at which the waveform propagates in space, and is found by dividing the wavelength by the period.
- ☐ B) The ratio between wave height and wavelength, indicating the steepness of the wave
- ☐ C) The time taken for one wavelength to pass through an arbitrary fixed point
- ☐ D) The ratio between the wave's wavelength and the water depth, indicating whether the wave is within the deep-water limit
- ☐ E) The ratio between wave height and angular frequency

Q 5-2) What is the wave frequency (f), period (T), angular frequency (ω), phase velocity (c) of a deep-water wave with a wavelength, λ , of 142 meters? Assume a fluid density of 1025 kg/m^3 and an acceleration due to gravity of 9.81 m/s^2 .

- ☐ A) $f = 0.069 \text{ Hz}$; $T = 14.475 \text{ seconds}$; $\omega = 0.434 \text{ rad/s}$; $c = 9.810 \text{ m/s}$
- ☐ B) $f = 0.074 \text{ Hz}$; $T = 13.487 \text{ seconds}$; $\omega = 0.466 \text{ rad/s}$; $c = 21.057 \text{ m/s}$
- ☐ C) $f = 0.105 \text{ Hz}$; $T = 9.537 \text{ seconds}$; $\omega = 0.659 \text{ rad/s}$; $c = 14.890 \text{ m/s}$
- ☐ D) $f = 0.148 \text{ Hz}$; $T = 6.743 \text{ seconds}$; $\omega = 0.932 \text{ rad/s}$; $c = 10.529 \text{ m/s}$
- ☐ E) $f = 0.042 \text{ Hz}$; $T = 23.905 \text{ seconds}$; $\omega = 0.263 \text{ rad/s}$; $c = 5.940 \text{ m/s}$

Q 5-3) Consider two waves with wave heights $H_1=2.2\text{m}$ and $H_2= 7.6\text{m}$, and corresponding wavelengths $\lambda_1=142\text{m}$ and $\lambda_2=187\text{m}$. For each wave, calculate the total energy per unit wave width (E_{width}). Assume a fluid density of 1025kg/m^3 and an acceleration due to gravity of 9.81 m/s^2 . Select the correct answer.

- ☐ A) $E_{1_{\text{width}}} = 431.92\text{ kW/m}$; $E_{2_{\text{width}}} = 6788.00\text{ kW/m}$
- ☐ B) $E_{1_{\text{width}}} = 178.48\text{ kW/m}$; $E_{2_{\text{width}}} = 235.04\text{ kW/m}$
- ☐ C) $E_{1_{\text{width}}} = 3455.39\text{ kW/m}$; $E_{2_{\text{width}}} = 54303.98\text{ kW/m}$
- ☐ D) $E_{1_{\text{width}}} = 785.32\text{ kW/m}$; $E_{2_{\text{width}}} = 3572.63\text{ kW/m}$
- ☐ E) $E_{1_{\text{width}}} = 863.85\text{ kW/m}$; $E_{2_{\text{width}}} = 13576.00\text{ kW/m}$

Q 5-4) Extraction of tidal energy is typically done by exploiting:

- ☐ A) The energy in waves generated by tides in shallow waters
- ☐ B) The temperature gradient energy of incoming and out-going tides
- ☐ C) The difference in salinity of tides
- ☐ D) Potential energy and/or kinetic energy harvesting
- ☐ E) The elastic energy in tides as they are funneled through estuaries

Question 21

Q6-1) Below you will find five (5) ocean energy technologies. Which technology does NOT utilize wave energy?

- ☐ A) Wave dragon
- ☐ B) Floating power plant
- ☐ C) Aquabuoys
- ☐ D) Offshore wind farm
- ☐ E) Pelamis

Question 22

Q6-2) Hydro power is a well developed renewable energy technology.

Which statement describes best the current status of large hydro power plants with a dam?

- ☐ A) High investment, low operation cost and no environmental impact
- ☐ B) It is generating power at a high cost with no emissions
- ☐ C) Relatively high environmental impact (change of landscape) and very low cost of power
- ☐ D) Can be used to store electricity and has a low investment
- ☐ E) Low environmental impact and high cost of power

Question 23

Q6-3) When locating a wave energy plant the most important to consider is the following:

- ☐ A) Short distance to the grid
- ☐ B) Short distance to shore (easy to repair) and deep water
- ☐ C) Low environmental impact and shallow water
- ☐ D) Long lifetime and far from the coast so that it cannot be seen from the shore
- ☐ E) Good wave energy resource, short distance to the grid and low environmental impact

Question 24

Q6-4 You are appointed as a consultant to the Governments in Scotland and Denmark. You are asked to estimate the area needed to cover 50% electricity consumption by wave energy both in Denmark and in Scotland. In Denmark half of the wave energy plants are located in the North Sea and the rest in the inner Danish waters.

We assume that the electricity consumption of Scotland and Denmark are 36 and 32 TWh respectively (2017).

In order to satisfy 50% of the above electricity consumption by wave energy Denmark will need an area in the sea that is larger by a factor of X than Scotland needs (other conditions assumed equal):

X is:

- ☐ A) 1 - same area
- ☐ B) 3 to 5
- ☐ C) 5 to 7
- ☐ D) 7 to 9
- ☐ E) > 10

Spørgsmål 12

Q 4-1) Below you will find five (5) renewable energy technologies that can produce electricity. Which technology(-ies) can be used to store energy?

- ☐ A) Wave dragon
- ☐ B) Floating power plant
- ☐ C) Tidal energy plant
- ☐ D) Hydro power plant with dam
- ☐ E) Pelamis

Spørgsmål 13

Q 4-2) There are three (3) categories or types of wave energy plants. They are:

- ☐ A) Oscillating water column, overtopping devices, and oscillating bodies
- ☐ B) Francis turbine, oscillating bodies and OTEC
- ☐ C) Overtopping devices, floating power plant, and oscillating water column
- ☐ D) Oscillating bodies, run-of-river plant, Pelton turbine
- ☐ E) Floating power plant, overtopping devices and Kaplan turbine

Spørgsmål 14

Q 4-3) Hydro power is a well developed renewable energy technology. Which statement describes best the current status of run-of-river hydro power plants?

- ☐ A) High investment, low operation cost and no environmental impact
- ☐ B) It is generating power at a high cost with no emissions
- ☐ C) Very low cost of power, no emissions and low environmental impact
- ☐ D) Can be used to store electricity and has a low investment
- ☐ E) Low environmental impact and high cost of power

Spørgsmål 15

Q 4-4) We assume that the Minister of Climate and Energy of Scotland has put forward a target of covering 50% of the electricity consumption of Scotland by wave energy by 2040. You are appointed as a consultant and asked to estimate the area needed in the sea. The average wave resource around Scotland is 70 kW/m wave front and we assume the wave energy units each rated at 4 MW are 100 x 100 m in size and are installed in the sea in a square pattern at a distance between the units of 600 m. The annual electricity consumption of Scotland is 36 TWh (2017).

Scotland will need an area in the sea of approximately:

- ☐ A) < 100 km²
- ☐ B) 200-300 km²
- ☐ C) 500-600 km²
- ☐ D) 800-900 km²
- ☐ E) > 1000 km²

Lecture 7 (Thermodynamics and electrochemistry)

restart : with(LinearAlgebra) :

Q 3-1) What is the thermodynamic function that describes the total energy that is released from the process of combustion of methanol at constant pressure?

- ☐ A) The enthalpy of the reaction
- ☐ B) The enthalpy of formation of methanol
- ☐ C) The entropy of methanol
- ☐ D) The Gibbs free energy of the reaction
- ☐ E) The Gibbs free energy of formation of methanol

Q 3-2) Which of the following statements can be true for a reversible process?

- ☐ A) The process takes place at a temperature higher than in the surroundings and it results in heat dissipation in the surroundings
- ☐ B) The process is accompanied by a loss of capability to perform work
- ☐ C) Entropy is generated
- ☐ D) Entropy is not generated
- ☐ E) There is a net (decisive, resulting) driving force in one direction

Q 3-3) Which one of the following processes is impossible (not possible) in an isolated system

- ☐ A) Complete conversion of work to work
- ☐ B) Complete conversion of work to heat
- ☐ C) Complete conversion of heat to work
- ☐ D) Complete conversion of heat to heat
- ☐ E) Complete conversion of work to electricity

Q 3-4) The following enthalpies of formation can be found in a table: $\Delta H_f(\text{CH}_4) = -74.6 \text{ kJ mol}^{-1}$, $\Delta H_f(\text{O}_2) = 0 \text{ kJ mol}^{-1}$, $\Delta H_f(\text{CO}_2) = -393.5 \text{ kJ mol}^{-1}$, $\Delta H_f(\text{H}_2\text{O (l)}) = -285.8 \text{ kJ mol}^{-1}$. How much energy is released when methane is combusted?

- ☐ A) 74.6 kJ mol^{-1}
- ☐ B) $318.9 \text{ kJ mol}^{-1}$
- ☐ C) $393.5 \text{ kJ mol}^{-1}$
- ☐ D) $604.7 \text{ kJ mol}^{-1}$
- ☐ E) $890.5 \text{ kJ mol}^{-1}$

Question 29

Q 8-1) Which of the following processes is NOT performance of work?

- ☐ A) Compressing a gas
- ☐ B) Stirring a liquid
- ☐ C) Shaking a drink
- ☐ D) The process that follows adding an ice cube to a drink
- ☐ E) Passing an electrical current through an electrolyte

Question 30

Q 8-2) What is the theoretical basis for Carnot's law?

- ☐ A) Conservation of energy alone
- ☐ B) Conservation of entropy alone
- ☐ C) Conservation of heat alone
- ☐ D) Conservation of energy and entropy
- ☐ E) Conservation of heat and entropy

Question 31

Q 8-3) Which of the five reaction are expected to have the highest generation of entropy per mole reaction as written?

- ☐ A) $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$ (at 25 °C)
- ☐ B) $\text{CO}_2 + 4\text{H}_2 \rightarrow \text{CH}_4 + 2\text{H}_2\text{O}$ (at 300 °C)
- ☐ C) $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ (at 500 °C)
- ☐ D) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O (l)}$ (at 25 °C)
- ☐ E) $2\text{H}_2\text{O (l)} \rightarrow 2\text{H}_2 + \text{O}_2$ (at 25 °C)

Question 32

Q 8-4) What is the standard enthalpy of combustion of ethane gas (C_2H_6) (the higher heating value):

Enthalpies of formation are:

$\Delta h_f^\circ (\text{C}_2\text{H}_6)$: -84.0 kJ mol⁻¹

$\Delta h_f^\circ (\text{CO}_2)$: -393.5 kJ mol⁻¹

$\Delta h_f^\circ (\text{H}_2\text{O (l)})$: -285.8 kJ mol⁻¹

- ☐ A) -1560.4 kJ mol⁻¹
- ☐ B) -595.3 kJ mol⁻¹
- ☐ C) -84.0 kJ mol⁻¹
- ☐ D) -763.3 kJ mol⁻¹
- ☐ E) -393.5 kJ mol⁻¹

Question 33

Q 9-1) What is meant by the term "overvoltage"

- ☐ A) A thermodynamic property of an electrochemical reaction
- ☐ B) The magnitude that the anode potential is higher than the cathode potential
- ☐ C) The magnitude that the cathode potential is higher than the anode potential
- ☐ D) A kinetic property. A measure of electrode polarization when a current is drawn
- ☐ E) An always positive voltage added to the equilibrium voltage when a current is drawn

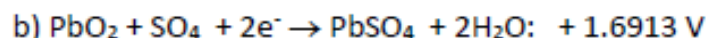
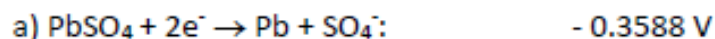
Question 34

Q 9-2) Which part of an electrochemical cell for energy conversion is typically causing the major part of the ohmic energy loss?

- ☐ A) The anode catalyst
- ☐ B) The cathode catalyst
- ☐ C) The electrolyte
- ☐ D) The sealing of the cell
- ☐ E) The connecting cables

Question 35

Q 9-3) In the lead acid battery, the standard reduction potentials for the electrode processes are the following:



What is the cell voltage of the battery based on this (*all activities = 1*)?

- ☐ A) 1.3325 V
- ☐ B) 1.6913 V
- ☐ C) 2.0501 V
- ☐ D) 2.665 V
- ☐ E) 4.1002 V

Question 36

Q 9-4) What is the Nernst equation used for?

- ☐ A) Calculation of overvoltage of one electrode
- ☐ B) Calculation of the combined overvoltage of two electrodes
- ☐ C) Calculation of electrode kinetics
- ☐ D) Calculation of equilibrium potentials
- ☐ E) Calculation of electrical efficiency

Spørgsmål 16

Q 5-1) Which one of the following quantities affects the reversible potential of an electrochemical cell?

- ☐ The resistance of the electrolyte
- ☐ The catalysts on the electrodes
- ☐ The concentration/pressure/activity of the chemical species taking part in the electrochemical reaction
- ☐ The current drawn from the cell
- ☐ The mass transport properties of the electrodes for gasses or liquids

Spørgsmål 17

Q 5-2) The maximum work obtainable from a chemical reaction is given by

- ☐ The enthalpy change of the reaction
- ☐ The free energy change of the reaction
- ☐ The enthalpy of formation
- ☐ The free energy of formation
- ☐ The entropy change of the reaction

Spørgsmål 18

Q 5-3) Can the electrical efficiency of an electrolyzer producing hydrogen and oxygen from water at room temperature be larger than 100 % based on thermodynamics?

- ☐ No, efficiencies can never be above 100 %
- ☐ Yes, the missing energy must then be supplied as heat
- ☐ Yes, the electrical energy input can be as small as one likes as long as the rest of the energy is supplied as heat
- ☐ No, it is always 100 % because energy is conserved (1st law)
- ☐ No, an electrolyzer works through processes, which are irreversible and the entropy production makes even 100 % efficiency impossible

Spørgsmål 19

Q 5-4) Which of the following statements describes the thermo-neutral cell voltage correctly?

- ☐ At the thermoneutral voltage, a water electrolyzer can be operated with no overall production or consumption of heat
- ☐ At the thermoneutral voltage, a hydrogen/air fuel cell can be operated with no overall production or consumption of heat
- ☐ The thermoneutral voltage is a thermodynamic abstraction and no real electrochemical cell can be operated at this voltage
- ☐ The thermoneutral voltage is the voltage at which the electrical energy input exactly matches the heat output of the cell
- ☐ The thermoneutral voltage is the voltage at which the electrical output exactly matches the heat input to the cell

Lecture 8 (Energy storage)

restart : with(LinearAlgebra) :

Question 25

Q 7-1) Rank the following energy storage technologies according to their efficiency (high to low)

- ☐ A) Flywheels > Pumped Hydro Storage (PHS) > flow batteries > Compress Air Energy Storage (CAES) > hydrogen
- ☐ B) Compress Air Energy Storage (CAES) > Pumped Hydro Storage (PHS) > flow batteries > hydrogen > flywheels
- ☐ C) Pumped Hydro Storage (PHS) > flow batteries > flywheels > hydrogen > Compress Air Energy Storage (CAES)
- ☐ D) Hydrogen > flywheels > Compress Air Energy Storage (CAES) > flow batteries > Pumped Hydro Storage (PHS)
- ☐ E) Pumped Hydro Storage (PHS) > flywheels > hydrogen > Compress Air Energy Storage (CAES) > flow batteries

Question 26

Q 7-2) BaCl_2 can be used for thermochemical energy storage as it can bind 8 NH_3 gas molecules with an enthalpy of reaction of 37.7 kJ/ (mol NH_3). How much energy must be supplied to release the 8 NH_3 molecules again (the molar mass of $\text{Ba}(\text{NH}_3)_8\text{Cl}_2$ is 344.48 g/mol)?

- ☐ A) 876 kJ/(g of $\text{Ba}(\text{NH}_3)_8\text{Cl}_2$)
- ☐ B) 109 kJ/(kg of $\text{Ba}(\text{NH}_3)_8\text{Cl}_2$)
- ☐ C) 876 J/(kg of $\text{Ba}(\text{NH}_3)_8\text{Cl}_2$)
- ☐ D) 243 Wh/(kg of $\text{Ba}(\text{NH}_3)_8\text{Cl}_2$)
- ☐ E) 30.4 Wh/(kg of $\text{Ba}(\text{NH}_3)_8\text{Cl}_2$)

Question 27

Q 7-3) How large was the fraction of thermal storage in new installations in utility-scale energy storage made in 2015 and 2016?

- ☐ A) <1%
- ☐ B) ~5%
- ☐ C) ~10%
- ☐ D) ~50%
- ☐ E) >90%

Question 28

Q 7-4) Rank the energy storage technologies according to their power density (low to high)

- ☐ A) Li-ion < flywheels < fuel cells < CAES < PHS
- ☐ B) CAES < fuel cells < PHS < Li-ion < flywheels
- ☐ C) Fuel cells < Li-ion < flywheels < PHS < CAES
- ☐ D) Flywheels < PHS < Li-ion < CAES < fuel cells
- ☐ E) PHS < CAES < fuel cells < flywheels < Li-ion

Spørgsmål 24

Q 7-1) Rank the following energy storage technologies according to their energy density (high to low)

- ☐ A) Compress Air Energy Storage (CAES) > Pumped Hydro Storage (PHS) > Na-S batteries > flywheels
- ☐ B) Na-S batteries > flywheels > Compress Air Energy Storage (CAES) > Pumped Hydro Storage (PHS)
- ☐ C) Pumped Hydro Storage (PHS) > flywheels > Compress Air Energy Storage (CAES) > Na-S batteries
- ☐ D) Na-S batteries > Pumped Hydro Storage (PHS) > flywheels > Compress Air Energy Storage (CAES)
- ☐ E) Flywheels > Compress Air Energy Storage (CAES) > Na-S batteries > Pumped Hydro Storage (PHS)

Spørgsmål 25

Q 7-2) The following statement about Phase Change Materials (PCM) is false:

- ☐ A) PCMs are substances that absorb and release thermal energy during e.g. melting and freezing
- ☐ B) Many different classes of materials can be PCMs
- ☐ C) A PCM can store different amounts of energy in its different phases
- ☐ D) PCMs typically operate at temperatures above 500 deg. C
- ☐ E) Water/ice is a commonly used PCM

Spørgsmål 26

Q 7-3) Rank the energy storage technologies according to their current levelized cost of storage (LCOS) (low to high)

- ☐ A) Na-S < thermo chemical < thermo latent heat < Power-2-Gas (H₂) < Pumped Hydro Storage (PHS)
- ☐ B) Thermo latent heat < Power-2-Gas (H₂) < Pumped Hydro Storage (PHS) < Na-S < thermo chemical
- ☐ C) Pumped Hydro Storage (PHS) < thermo chemical < Na-S < Power-2-Gas (H₂) < thermo latent heat
- ☐ D) Power-2-Gas (H₂) < Pumped Hydro Storage (PHS) < Na-S < thermo chemical < thermo latent heat
- ☐ E) Thermo chemical < Na-S < Pumped Hydro Storage (PHS) < thermo latent heat < Power-2-Gas (H₂)

Spørgsmål 27

Q 7-4) How much energy can 4 m³ of granite store if it is heated from 20 to 450 deg. C (specific heat: 774 J/kgK; density: 2700 kg/m³)?

- ☐ A) ~1 kWh
- ☐ B) ~1 MWh
- ☐ C) ~1 kJ
- ☐ D) ~1 MJ
- ☐ E) ~1 GJ

▼ Lecture 9 (Fuel cells)

restart : with(*LinearAlgebra*) :

Q 9-1) What is the desired role of the electrolyte in a fuel cell

- ☐ A) Storage of reactants
- ☐ B) Conduction of electrons
- ☐ C) Conduction of ions
- ☐ D) Dissolution of hydrogen
- ☐ E) Dissolution of oxygen

Q 9-2) What is the role of the catalyst(s) in a fuel cell?

- ☐ A) It directs the reactants to the electrolyte where the conversion process takes place
- ☐ B) It is consumed by the electrochemical process
- ☐ C) It is only there because it is a good electronic conductor
- ☐ D) It makes the electrochemical process proceed more easily
- ☐ E) Makes an electrochemical process proceed faster

Q 9-3) What is the usual oxygen catalyst material in a PEM fuel cell?

- ☐ A) Platinum
- ☐ B) Iridium
- ☐ C) Mixed metal oxides
- ☐ D) Carbon
- ☐ E) Nafion

Q 9-4) What limits the operating temperature of a PEM fuel cell, practically below 90 °C?

- ☐ A) The electrolyte membrane dries at temperatures higher than 90 °C and loses the ion conductivity
- ☐ B) The electrolyte membrane melts at temperatures higher than 90 °C
- ☐ C) The catalyst loses activity at temperature higher 90 °C
- ☐ D) Hydrogen and oxygen reacts spontaneously at temperatures higher 90 °C
- ☐ E) The electrochemical process cannot be controlled at temperatures higher 90 °C

Question 37

Q 10-1) Why is the electrical efficiency of a fuel cell not limited by Carnot's Law?

- ☐ A) A fuel cell does not produce heat at all, and therefore Carnot's Law cannot be applied
- ☐ B) A fuel cell produces electricity directly and not via a heat to power process. It is not a heat engine
- ☐ C) A fuel cell is heat consuming and this would lead to a negative efficiency if Carnot's law were applied
- ☐ D) Thermodynamics have developed since Carnot. He could not foresee the fuel cell and therefore his law does not address fuel cells
- ☐ E) A fuel cell is indeed limited by Carnot's Law. Carnot's Law is based on thermodynamics and is therefore always the ultimate limit

Question 38

Q 10-2) The size of the standard free energy of the reaction $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ decreases with temperature. This causes a corresponding lowering of the equilibrium cell voltage of hydrogen powered fuel cells. Why is a solid oxide fuel cell at 800 °C not less efficient than the low temperature fuel cells at 80 °C?

- ☐ A) The individual cell is actually much less efficient, but the cells are thinner and more cells are applied in the stack
- ☐ B) The electrons contain much more energy at high temperature
- ☐ C) The high temperature lead to higher gas pressures (the ideal gas law) and consequently higher activities, which lead to a higher cell voltage
- ☐ D) Carnot's law compensate for the low free energy (higher temperature, higher efficiency)
- ☐ E) The electrode kinetics are much faster at high temperature leading to a lower overvoltage

Question 39

Q 10-3) Which one of the following arguments for converting hydrogen into carbon containing synthetic fuels is NOT sound?

- ☐ A) The process consumes CO_2 that would otherwise be emitted to the surroundings. Thus, it has a long-term positive effect on the global warming
- ☐ B) Biogas contains a significant amount of CO_2 from the fermentation process. It can increase the value of the biogas if it is upgraded by methanation of the CO_2
- ☐ C) Fuels like these are already common in our energy system and can therefore be used directly in the present energy infrastructure
- ☐ D) Synthetic fuels are generally easier or less expensive to store than hydrogen
- ☐ E) Carbon containing fuels are more energy dense than hydrogen

Question 40

Q 10-4) The PEM electrolyzer has become very popular in recent years. Why is that?

- ☐ A) It is very compact, due to high current density at a reasonable cell voltage
- ☐ B) It is less expensive than any other type of electrolyzer on the market
- ☐ C) It is the only well-proven system on the market
- ☐ D) It can be manufactured without the use of noble metals
- ☐ E) The fundamental thermodynamics are more attractive for the PEM electrolyser than for other types

Spørgsmål 20

Q 6-1) If fuel cells are compared to thermally driven engines like combustion engines and turbines, then

- ☐ fuel cells are always more efficient because of Carnot's law
- ☐ fuel cells are typically more efficient because of Carnot's law, but it also depends on the overvoltages
- ☐ fuel cells are more efficient because they don't produce heat
- ☐ fuel cells are more efficient because they are operated at low temperature
- ☐ fuel cells are more efficient because they don't produce any entropy

Spørgsmål 21

Q 6-2) A fuel cell is operated at 0.65 V with pure hydrogen and pure oxygen, both at 1 bar. The thermo-neutral voltage is 1.48 V and the reversible voltage is 1.23 V. What is the electrical efficiency with reference to the higher heating value (HHV)?

- ☐ 25 %
- ☐ 44 %
- ☐ 53 %
- ☐ 58 %
- ☐ 83 %

Spørgsmål 22

Q 6-3) Which factor is important for the ion conducting membrane in a PEMFC

- ☐ Very dry operating conditions to avoid the membrane becoming too soft
- ☐ Temperatures above 100 °C to secure that the water produces is vapour
- ☐ Low water content in the membrane to provide high proton conductivity
- ☐ High water content in the membrane to provide high proton conductivity
- ☐ The membrane must have sufficiently large pores to let the electrolyte to flow through

Spørgsmål 23

Q 6-4) Today, hydrogen is stored in fuel cell cars as

- ☐ Compressed hydrogen at 200 bar
- ☐ Compressed hydrogen at 700 bar
- ☐ Liquid hydrogen
- ☐ Metal hydrides
- ☐ Methanol

▼ Lecture 10 (Batteries)

restart : with(LinearAlgebra) :

Q 10-1) What is the theoretical capacity of a typical positive electrode material in a Li-ion batter

- ☐ A) ~200 Wh/L
- ☐ B) ~200 Wh/kg
- ☐ C) ~200 kWh/kg
- ☐ D) ~200 mAh/g
- ☐ E) ~200 Ah/g

Q 10-2) Arrange the following battery chemistries according to their gravimetric energy density (energy weight), from lowest to highest:

- ☐ A) Ni-MH < Pb-acid < Li-ion < Ni-Cd
- ☐ B) Pb-acid < Li-ion < Ni-MH < Ni-Cd
- ☐ C) Li-ion < Ni-MH < Ni-Cd < Pb-acid
- ☐ D) Pb-acid < Ni-Cd < Ni-MH < Li-ion
- ☐ E) Ni-Cd < Li-ion < Pb-acid < Ni-MH

Q 10-3) If a battery with a Coulombic Efficiency (sometimes called "cycle efficiency") of 99.9 % is charged and discharge once a day, how much of the original capacity is left in the battery after one year?

- ☐ A) 99.6 %
- ☐ B) 96.4 %
- ☐ C) 69.4 %
- ☐ D) 2.5 %
- ☐ E) 0.0 %

Q 10-4) The following overall reaction takes place in a lithium-iodine battery: $\text{Li} + \frac{1}{2} \text{I}_2 \rightarrow \text{LiI(s)}$. Which component is oxidized and which is reduced?

- ☐ A) Lithium is reduced and Iodine is oxidized
- ☐ B) Lithium is oxidized and Iodine is reduced
- ☐ C) Both Lithium and Iodine are oxidized
- ☐ D) Both Lithium and Iodine are reduced
- ☐ E) Neither Lithium nor Iodine is reduced or oxidized

Question 41

Q 11-1) How much energy can be stored in a battery with a capacity of 3000 C and a cell voltage of 3.6 V?

- ☐ A) 10.8 Wh
- ☐ B) 3.0 Wh
- ☐ C) 10.8 W
- ☐ D) 3.0 W
- ☐ E) 10.8 kWh

Question 42

Q 11-2) The following statement is true for Vanadium Redox Flow Batteries:

- ☐ A) The accessible energy and power can be scaled independently & the cost of vanadium and the cycle-life are both low
- ☐ B) The accessible energy and power cannot be scaled independently & the cost of vanadium and the cycle-life are both low
- ☐ C) The accessible energy and power can be scaled independently & the cost of vanadium and the cycle-life are both high
- ☐ D) The accessible energy and power cannot be scaled independently & the cost of vanadium and the cycle-life are both high
- ☐ E) The accessible energy and power can be scaled independently & the cost of vanadium is high and the cycle-life is low

Question 43

Q 11-3) The following reaction takes place in a sodium-sulphur battery: $4S + 4Na \rightarrow 4Na_2S$. Which component is oxidized and which is reduced?

- ☐ A) Neither sodium nor sulphur is reduced or oxidized
- ☐ B) Both sulphur and sodium are reduced
- ☐ C) Both sulphur and sodium are oxidised
- ☐ D) Sulphur is oxidized and sodium is reduced
- ☐ E) Sulphur is reduced and sodium is oxidized

Question 44

Q 11-4) A battery with a capacity of 3000 mAh is discharged at a C-rate of 2C. How long time does it take to discharge the battery fully?

- ☐ A) 180 min
- ☐ B) 120 min
- ☐ C) 60 min
- ☐ D) 30 min
- ☐ E) 15 min

Spørgsmål 28

Q 8-1) Arrange the following battery chemistries according to their gravimetric energy density (energy per weight), from highest to lowest

- ☐ A) Ni-MH > Pb-acid > Li-ion > Ni-Cd
- ☐ B) Pb-acid > Li-ion > Ni-MH > Ni-Cd
- ☐ C) Li-ion > Ni-MH > Ni-Cd > Pb-acid
- ☐ D) Pb-acid > Ni-Cd > Ni-MH > Li-ion
- ☐ E) Li-ion > Ni-Cd > Pb-acid > Ni-MH

Spørgsmål 29

Q 8-2) What is the theoretical capacity of a graphite electrode material in a Li-ion battery

- ☐ A) ~400 kWh/L
- ☐ B) ~400 Wh/kg
- ☐ C) ~400 kWh/kg
- ☐ D) ~400 mAh/g
- ☐ E) ~4000 mAh/g

Spørgsmål 30

Q 8-3) A battery with a capacity of 10.000 C is charged at a C-rate of 1C. How long time does it take to charge the battery fully?

- ☐ A) 1 sec
- ☐ B) 10 sec
- ☐ C) 1 min
- ☐ D) 1 h
- ☐ E) 10.000 sec

Spørgsmål 31

Q 8-4) The following reactions take place in a rechargeable alkaline zinc-air (Zn-O₂) battery. Negative electrode: $\text{Zn} + 4\text{OH}^- \leftrightarrow \text{ZnO} + \text{H}_2\text{O} + 2\text{OH}^-$; Positive electrode: $\text{O}_2 + \text{H}_2\text{O} + 4\text{e}^- \leftrightarrow 4\text{OH}^-$; Overall: $2\text{Zn} + \text{O}_2 \leftrightarrow 2\text{ZnO}$. Which component is being reduced and which is oxidized during charging of the battery?

- ☐ A) Neither zinc nor oxygen is reduced or oxidized
- ☐ B) Both oxygen and zinc are reduced
- ☐ C) Both zinc and oxygen are oxidized
- ☐ D) Oxygen is oxidized and zinc is reduced
- ☐ E) Zinc is reduced and oxygen is oxidized

Lecture 11 (Electrolysis and syn fuels)

restart : with(LinearAlgebra) :

Q 11-1) Which type of electrolyzer depends on noble metal based catalysts?

- ☐ A) The solid oxide electrolyzer, since only noble metals are stable at the high temperature
- ☐ B) The alkaline electrolyzer, because the alkaline electrolyte is so corrosive
- ☐ C) The PEM electrolyzer, because the environment is acidic due to the sulfonic acid
- ☐ D) All the three types. Only noble metal catalysts are active
- ☐ E) None of the systems need noble metals for catalysts

Q 11-2) What is the maximum theoretical electrical efficiency of an electrolyzer cell with respect to the higher heating value, HHV, ie. With water fed as liquid (based only on thermodynamics at room temperature)?

- ☐ A) There is no maximum. The electrolyzer is not limited by thermodynamics
- ☐ B) ca. 50 % half
- ☐ C) ca. 80% because heat must be evolved
- ☐ D) 100 % because it is not limited by the Carnot maximum
- ☐ E) 121 % because thermal energy can be included

Q 11-3) What is the advantage of a zero-gap configuration

- ☐ A) Gas leaks are minimized
- ☐ B) The internal resistance of the electrolyte is small
- ☐ C) Heat is conducted more efficiently
- ☐ D) The catalysts work better
- ☐ E) Less electrolyte is needed

Q 11-4) What is the "thermo-neutral voltage"?

- ☐ A) Any voltage at which increased temperature does not affect electrode kinetics
- ☐ B) The voltage at which increased temperature does not increase the conversion rate
- ☐ C) A voltage that is not affected by temperature
- ☐ D) The voltage at which the electrolysis process can be driven by heat alone
- ☐ E) The voltage, 1.48 V, at which the heat produced by internal losses exactly matches the heat demand of the electrolysis process

Question 39

Q 10-3) Which one of the following arguments for converting hydrogen into carbon containing synthetic fuels is NOT sound?

- ☐ A) The process consumes CO₂ that would otherwise be emitted to the surroundings. Thus, it has a long-term positive effect on the global warming
- ☐ B) Biogas contains a significant amount of CO₂ from the fermentation process. It can increase the value of the biogas if it is upgraded by methanation of the CO₂
- ☐ C) Fuels like these are already common in our energy system and can therefore be used directly in the present energy infrastructure
- ☐ D) Synthetic fuels are generally easier or less expensive to store than hydrogen
- ☐ E) Carbon containing fuels are more energy dense than hydrogen

Question 40

Q 10-4) The PEM electrolyzer has become very popular in recent years. Why is that?

- ☐ A) It is very compact, due to high current density at a reasonable cell voltage
- ☐ B) It is less expensive than any other type of electrolyzer on the market
- ☐ C) It is the only well-proven system on the market
- ☐ D) It can be manufactured without the use of noble metals
- ☐ E) The fundamental thermodynamics are more attractive for the PEM electrolyser than for other types

Spørgsmål 40

Q 11-1) Which type of electrolyzer is known for a gap configuration (in contrast to zero gap)?

- ☐ The traditional alkaline electrolyzer
- ☐ The PEM electrolyzer
- ☐ The solid oxide electrolyzer
- ☐ All three types
- ☐ None of them

Spørgsmål 41

Q 11-2) Why are solid oxide electrolyzers always operated at high temperature (typically 700-900 °C)?

- ☐ Some of the metal oxide ceramic materials they are made from cannot tolerate liquid water (they dissolve)
- ☐ The electrolyte must be molten before it can conduct the oxygen ions
- ☐ The high temperature is necessary for cooling the cells
- ☐ It is the easiest way to evaporate water and then it takes less energy to electrolyze water vapour
- ☐ The electrolyte is a solid metal oxide that conducts oxygen ions. The ion movement between the position in the structure requires high thermal energy

Spørgsmål 42

Q 11-3) Methane can be produced by the Sabatier process from H_2 and CO_2 and this way, large amounts of energy can be stored in the natural gas grid and underground caverns. Which one of the following arguments against this process is correct?

- ☐ The energy efficiency is limited. The reaction enthalpy is negative and thus the process is exothermic (waste heat producing). Besides, energy is needed to collect and concentrate CO_2 from a suitable source
- ☐ The enthalpy of reaction is highly negative and thus a large-scale high temperature heat source is needed for the process. This may not be available
- ☐ When the stored methane is used later, it releases CO_2 and therefore it is just as harmful to the climate as fossil natural gas
- ☐ Methane is flammable and flammable fuels should always be avoided for safety reasons
- ☐ There are no arguments against making methane by the Sabatier process

Spørgsmål 43

Q 11-4) Rank hydrogen, synfuels (methanol, methane etc.) and batteries in terms of practical energy density (i.e. including tank/casing) for medium and large-scale application (cars, trucks ships)

- ☐ Hydrogen > synfuels > batteries
- ☐ Hydrogen > batteries > synfuels
- ☐ Synfuels > batteries > hydrogen
- ☐ Synfuels > hydrogen > batteries
- ☐ Batteries > synfuels > hydrogen

▼ Lecture 12 (Infrastructure and cost)

restart : with(LinearAlgebra) :

Q 12-1) If you increase the Weighted Average Cost of Capital (WACC) for a given project, the Net Present Value (NPV) goes:

- ☐ A) To zero
- ☐ B) Down
- ☐ C) Up
- ☐ D) No change (the WACC doesn't affect the NPV)
- ☐ E) To infinity

Q 12-2) The LCOE is used to:

- ☐ A) Compare the reliability different energy sources on an equal basis
- ☐ B) Compare the construction costs of different energy sources on an equal basis
- ☐ C) Compare the Life-Cycle Operational Environmental impact of different energy sources
- ☐ D) Compare the running costs (OPEX) of different energy sources on an equal basis
- ☐ E) Compare the cost of produced energy on an equal basis

Q 12-3) All the car engines of the World running at full power would generate a combined power of roughly:

- ☐ A) ~1 TW
- ☐ B) ~10 TW
- ☐ C) ~100 TW
- ☐ D) ~1.000 TW
- ☐ E) ~10.000 TW

Q 12-4) If all the cars in Denmark were electric and participated in a V2G (Vehicle-to-Grid) program they would in practice be able to run the grid for a time period on the order of (give or take a factor of 3):

- ☐ A) ~20 min.
- ☐ B) ~1 hour
- ☐ C) ~4 hours
- ☐ D) ~1 day
- ☐ E) ~1 week

Question 45

Q 12-1) If the cost of equity is much higher than the cost of debt (i.e. the shareholders demand much higher dividends than the bondholders demand interest) why would a business still chose some equity financing despite giving a higher WACC (weighted average cost of capital)?

- ☐ A) To make shareholders happy
- ☐ B) In order to decrease leverage
- ☐ C) Due to legal requirements
- ☐ D) Because the WACC doesn't matter when raising capital
- ☐ E) Because the WACC, although important, shouldn't be too low

Question 46

Q 12-2) Why is the DCF (discounted cash flow) of a future revenue stream less than the raw sum of that future revenue stream?

- ☐ A) Because DCF discounts future cash flow by the assumed business risks
- ☐ B) Because DCF discounts future cash flow by the inflation rate
- ☐ C) Because DCF discounts future cash flow by the cost of capital
- ☐ D) Because DFC discounts CAPEX by OPEX
- ☐ E) Because DFC adjusts the WACC over time

Question 47

Q 12-3) How much higher is the capacity factor of nuclear power compared to photovoltaic power, roughly?

- ☐ A) ~ 100 times higher
- ☐ B) ~ 15 times higher
- ☐ C) ~ 4 times higher
- ☐ D) ~ 1.5 times higher
- ☐ E) roughly the same

Question 48

Q 12-4) Denmark consumes around 3.5 GW of electricity on average. Roughly, how many typical family car engines (running at full power) does that power level correspond to?

- ☐ A) ~50.000 cars (2% of the Danish car fleet)
- ☐ B) ~250.000 cars (10% of the Danish car fleet)
- ☐ C) ~850.000 cars (34% of the Danish car fleet)
- ☐ D) ~1.300.000 cars (52% of the Danish car fleet)
- ☐ E) ~5.000.000 cars (200% of the Danish car fleet)

Spørgsmål 44

Q 12-1) The LCOE compares different electricity sources on a (discounted) total cost basis, but what does it not take into account.

- ☐ A) Externalities and dispatchability costs
- ☐ B) Maintenance costs
- ☐ C) Financing costs
- ☐ D) Procurement and contracting costs
- ☐ E) Permitting costs

Spørgsmål 45

Q 12-2) What is the capacity factor?

- ☐ A) The ratio of renewable energy to total energy in the grid
- ☐ B) The ratio of CO₂ emissions of two different sources of power - such as natural gas or coal
- ☐ C) The ratio of CAPEX to OPEX of a project
- ☐ D) The ratio of annual average power produced to the ideal case power
- ☐ E) The ratio of annual average power produced to the installed cost

Spørgsmål 46

Q 12-3) If you increase the lifetime (n) of an electricity generating installation - say from 30 years to 31 years, while keeping everything else constant (i.e. interest rates, fuel costs, maintenance costs, etc.) how does this effect the LCOE?

- ☐ A) LCOE is unaffected
- ☐ B) LCOE goes up substantially because the gain is not discounted many years ahead
- ☐ C) LCOE goes up goes up slightly because the gain is strongly discounted
- ☐ D) LCOE goes down substantially because the gain is not discounted many years ahead
- ☐ E) LCOE goes up goes down slightly because the gain is strongly discounted

Spørgsmål 47

Q 12-4) The global annual production of nickel (Ni) is about 1.25 billion kg. If we assume a car battery contains 0.75 kg/kWh, how many year's production of nickel would be required to convert the global car fleet to battery-electric vehicles? Assume an average car battery size of 70 kWh

- ☐ A) 0.1 years of global Ni production
- ☐ B) 2.5 years of global Ni production)
- ☐ C) 10 years of global Ni production
- ☐ D) 45 years of global Ni production
- ☐ E) 130 years of global Ni production